Vocabulary resonance in US trademark filings:
A measurement construct for commercial novelty, with validation and applications

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Abstract

The DeDeo prospective/retrospective Kullback–Leibler resonance framework (Barron et al., 2018; Murdock et al., 2017) was developed on cognitive corpora in which the author of each document is the agent whose novelty is being measured: Darwin's reading notebooks, parliamentary speeches of the French Revolution. This paper applies the framework to a qualitatively different corpus type, the goods/services descriptions of US trademark filings, and introduces the resulting construct, denoted ΔKL, as a firm-level measure of commercial novelty that is cheap to compute, available for millions of firms that never patent, and empirically distinct from patent-track invention.

Three validation results anchor the construct. First, ΔKL behaves as a marker of commercial risk-taking, with opposite signs at the mark and firm levels. Forward-leaning marks fare worse in the legal lifecycle: they register slightly less often, and among 2016–2018 registrations facing their first Section 8 maintenance gate, survival falls from 49.2% to 43.3% across ΔKL quintiles (n = 765,154; negative in 31 of 41 industries). The firms that file them fare better: eventual public listing among first-time filers rises ∼40% in relative terms across ΔKL quintiles, gross margin on an SEC-matched panel runs +3.2pp per σ ($t = 5.3$), and excess stock returns run +1.5 to +5.0pp per σ at 1–4 year horizons, decaying by year five. Prospective surprise alone predicts none of the firm-level outcomes; only the signed resonance does. Second, ΔKL is orthogonal-to-negative with patent counts within firm (-0.048σ, $t = -7.9$ on 14,470 matched firms), while expert "most innovative" lists sit $+0.34\sigma$ above the panel mean ($p < 0.001$); the construct captures a market-facing dimension of novelty that patents miss. Third, the construct supports direct measurement of cross-industry idea diffusion: software- and tech-services-origin themes arrive in transport and advertising/retail with 1–13 year lags, theme adoption follows Bass curves with median $q/p \approx 6.5$, and the cross-class flow is strongly asymmetric (software originates 33 of the 72 origin-to-arrival edges and receives only 10).

The paper closes with a research agenda the construct opens: whether ideas that transfer between industries hold their value, whether suppliers to a new theme outperform its deployers, and how demographic concentration within industries relates to where new themes originate. Code and firm-year panels are public at `https://github.com/ericsilver/novelty`.

1 Introduction

Innovation measurement leans heavily on patents. Patent-based measures are well-validated for technical invention but cover a narrow slice of commercial novelty: most firms never patent,

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service-sector and business-model innovation rarely patents, and patenting propensity varies so much across industries that cross-industry comparisons are fraught (Dinlersoz et al., 2018; Graham et al., 2013). Trademark filings offer a complementary window. Nearly every commercially serious product or service introduction in the United States is accompanied by a trademark filing, the filing's goods/services description states in plain commercial language what the firm intends to sell, and the corpus extends across every sector of the economy at the rate of hundreds of thousands of granted filings per year. The text of those descriptions, however, has been largely unused; the literature has worked with filing counts and filing metadata rather than filing language.

Murdock et al. (2017) introduced a measurement framework for directional novelty in time-ordered text corpora: for each document, compare its token distribution to the aggregated distribution of past documents (prospective surprise) and future documents (retrospective surprise). The signed difference of those two quantities they call *resonance*: it is positive when a document is unusual against its recent past and close to its near future, negative for the inverse. The framework was developed on Charles Darwin's reading notebooks and extended in Barron et al. (2018) to French Revolution parliamentary debates. Both corpora share a salient property: the text is authored by the agent whose novelty is the object of measurement.

This paper applies the same apparatus to trademark goods/services text, which is commercially incentivised, typically drafted by counsel, and exists to define legal protection scope rather than to express cognition. Whether the resonance framework produces interpretable structure on text of this kind is the threshold question, and the answer this paper documents is yes. The construct, denoted ΔKL, behaves as a marker of commercial risk-taking, with opposite signs at the mark and firm levels. Filings whose vocabulary leans toward where their industry's language is going fare worse as legal objects: they register slightly less often, and the registered marks fail the first Section 8 maintenance gate more often. The firms that file them fare better as businesses: higher gross margins, excess stock returns at one-to-four-year horizons, and a higher rate of eventually going public among first-time filers. The same signal is uncorrelated-to-negatively correlated with patenting within firm, while firms on expert "most innovative" lists score high on it; ΔKL captures a market-facing dimension of novelty that the patent record does not.

Because the construct is computed from public text with public code, it scales to questions that firm-survey or patent-linked measures cannot easily reach. The paper demonstrates one such application in depth, measuring the diffusion of business vocabulary across industries directly at the phrase and theme levels, and sketches a research agenda: the portability of transferred ideas, the relative fortunes of suppliers versus deployers of a new theme, and the relationship between demographic concentration within an industry and where new themes are born.

Section 2 defines the construct, the corpus, and known fragilities. Section 3 reports the commercial-outcome validation. Section 4 reports the within-firm patent contrast and the expert-list discriminant. Section 5 reports the cross-industry diffusion application. Section 6 reports a base-rate discipline result on entrant-vs-incumbent classification. Section 7 states the research agenda. Section 8 bounds the claims. Section 9 is reproducibility.

2 The construct

2.1 The DeDeo apparatus on trademark text

For a time-ordered sequence of documents, with each document at year t producing a token distribution P_t, the framework defines two quantities against a window W of past and future documents (Barron et al., 2018; Murdock et al., 2017):

- *Prospective surprise*: $\mathrm{KL}(P_t \,\|\, Q_{\text{past}})$, where Q_{past} is the aggregate token distribution of documents in years $t-W$ to $t-1$.
- *Retrospective surprise*: $\mathrm{KL}(P_t \,\|\, Q_{\text{future}})$, where Q_{future} is the aggregate distribution of documents in years $t+1$ to $t+W$.

The signed resonance is

$$\Delta\mathrm{KL}_t \;=\; \mathrm{KL}(P_t \,\|\, Q_{\text{past}}) - \mathrm{KL}(P_t \,\|\, Q_{\text{future}}).$$

Positive ΔKL marks a document whose vocabulary is unusual relative to its recent past and close to its near future: the field moved toward this filing. Negative ΔKL marks the inverse, language that looked normal at filing and looked played-out in hindsight: the field moved away.

2.2 Corpus and conditioning

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. The deepest diagnostic work in this paper uses the complete software/electronics class (NICE 009; 1.81M filings), and the diffusion application uses four classes selected to exercise canonical cross-industry flow: software (009), scientific and technical services (042), transport (039), and advertising and retail (035), totalling 2.32 million granted filings 1990–2024. The firm-level panels link trademark owners to SEC EDGAR financial filers and to PatentsView patent assignees.

Two conditioning rules apply throughout. The corpus is restricted to *granted* filings (registration date populated), screening out speculative or defensive applications that did not survive examination. And the lifecycle outcomes in Section 3 keep the examination step and the maintenance step separate: registering and maintaining are different commercial acts with different drivers, and pooling them conflates examiner behavior with the use requirement. Mark status is read from the USPTO status-code dictionary directly (700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired), and maintenance is measured on registration cohorts for which exactly one Section 8 gate has elapsed. The dictionary-and-cohort design replaces a heuristic outcome coding used in earlier iterations of this analysis chain; the data provide a status snapshot rather than a status history, so the only way to attribute a cancellation to a specific maintenance gate is to restrict to cohorts in which exactly one gate has passed, and the only validated map from snapshot to outcome is the official code dictionary, ground-truthed here against marks with publicly known fates.

2.3 Computation

For each granted filing at year t in class c with goods/services text producing token distribution P (unigram + bigram, after a standard analyzer that drops stopwords), the reference distributions Q_{past} and Q_{future} are computed from all granted filings in class c in the respective windows. I use $W = 5$ years (flat window) with Dirichlet smoothing ($\alpha = 1$ per cell). The within-class vocabulary is the set of unigrams and bigrams with document frequency ≥ 50.

The signal depends on having enough past and future filings to populate the references. Figure 1 shows the reference-window density by year for two diagnostic classes; the 5-year past window is incomplete before 1990 and the 5-year future window is incomplete after 2020, and both zones are excluded from the construct's interpretable range. Within the supported window the classes analyzed here have between ∼3,500 and ∼80,000 filings per year.

The burn-in cut is set empirically rather than by the window arithmetic alone. Thin past references inflate prospective surprise, so contamination appears as a positive bias in the per-year mean of signed ΔKL relative to the class's long-run level. Profiling that bias across all 44 classes with sufficient volume: the median absolute bias is ∼2.0 nats in 1990, 0.25 in 1991, 0.10 in 1992, and 0.04 from 1993 onward, below any effect size this paper interprets. The deviations that reappear in 1995–1997 (∼0.15–0.17) are not burn-in, because mechanical contamination cannot rebound; they are the dot-com era moving the corpus, which a longer burn-in would discard as data. Per-class burn-in lengths were also estimated under a stability-band criterion; the apparent cross-class differences track each class's noise floor rather than its reference density (correlation of burn-in length with early filing volume +0.26, with a slow-moving-class proxy +0.14), so there is no empirical case for industry-specific burn-ins. All analyses therefore use a single standard cut at 1995, the first scored year whose full 5-year past window lies inside the dense corpus, which the bias profile shows is conservative by about two years.

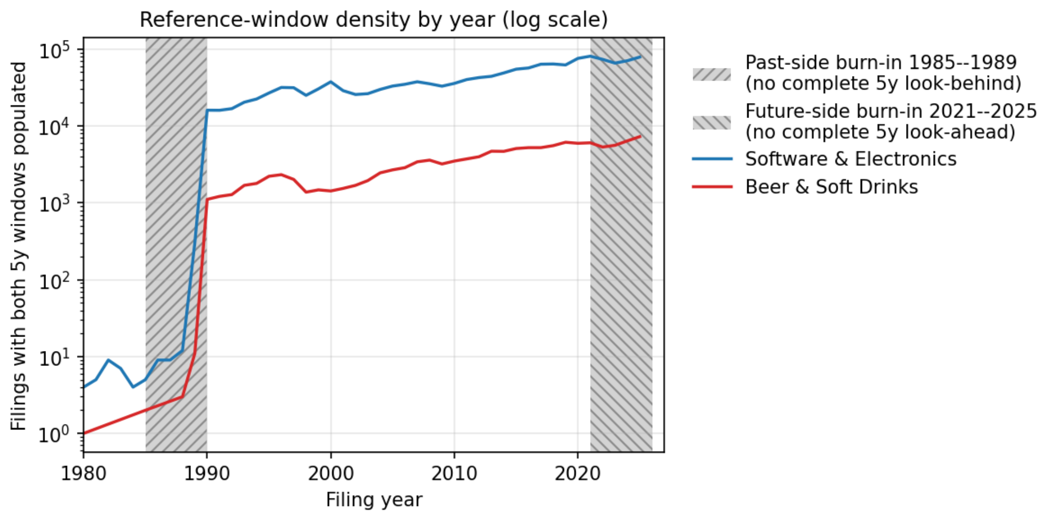

Figure 1: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete.

2.4 The construct at the case level

Figure 2 grounds the construct in recognizable examples. The two scatter axes are prospective surprise and retrospective surprise, with ΔKL as the diagonal residual. Filings below the diagonal are *innovator*: the language looked unusual at filing and became standard afterwards. Filings above the diagonal are *tired*: the past resembles the filing more than the future does.

2.5 Known fragilities

The construct has documented fragilities on this kind of text. On the complete Class 009 corpus, prospective and retrospective KL are correlated at $r = +0.971$ across 1.56M scored filings; signed ΔKL is therefore a small residual of two large, highly correlated quantities, and the residual is noisy at the per-filing level. The magnitude |ΔKL| inflates monotonically for short filings (0.293 for documents under 5 terms, decaying to 0.147 for documents of 40 or more terms), a length artifact that requires control in any per-filing analysis. Pre-1995 readings are contaminated by thin past

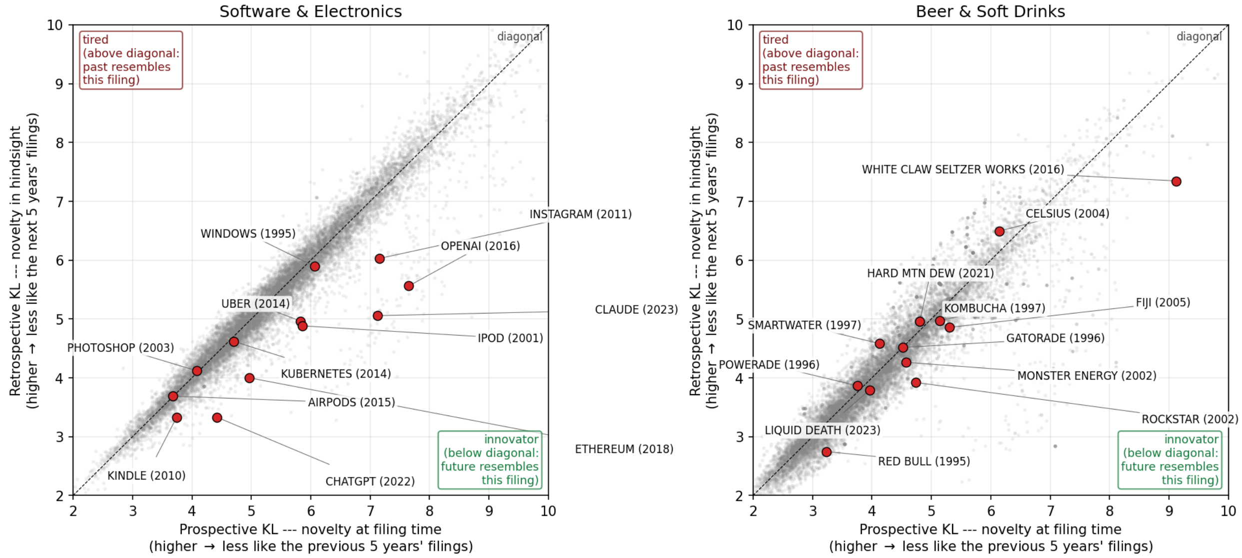

Figure 2: Prospective KL versus retrospective KL with ΔKL as the above-diagonal/below-diagonal residual. Labeled examples are recognizable post-burn-in brand introductions, positioned by their goods/services text. *Left:* software/electronics. OpenAI (+2.09), Claude (+2.06), Instagram, ChatGPT, iPod, Ethereum, and Uber sit on the innovator side; Photoshop (2003) and AirPods sit essentially on the diagonal. *Right:* a beer/soft-drinks diagnostic class. White Claw (+1.78), Rockstar, and Red Bull lean innovator; Smartwater (1997) and Celsius (2004) lean tired.

references. Window sensitivity is bounded: across $W \in \{3, 5, 7\}$ years, 6.9–10.4% of filings flip sign, well under conventional fragility thresholds. The per-filing noise attenuates at every aggregation used in this paper: firm-year means, token-level pools, and theme-level prevalence trajectories.

A topic-distribution variant addresses the short-document concern directly, following the operationalization of the framework's source papers, which scored Darwin's notebooks and the French Revolution debates on topic distributions rather than raw token distributions (Barron et al., 2018; Murdock et al., 2017). Re-scoring each filing's prospective and retrospective KL on its 50-topic LDA distribution (the topic model of Section 5) gives every filing a dense representation regardless of token count, and the length artifact largely dissolves: mean $|\Delta\text{KL}|$ varies 2.5-fold across document-size bands under token scoring but only $\sim$1.4-fold under topic scoring. The two scorings agree moderately per filing (Pearson 0.46, Spearman 0.36 on 1.9M jointly scored filing-class observations), and the prospective/retrospective correlation is nearly as high on topics (0.950) as on tokens (0.980), consistent with the stability reading: what has not been seen before will mostly not be seen again, and the signal lives in the difference. Re-estimating the outcome analyses under topic scoring is the natural robustness pass and is underway as follow-up work.

3 Validation against commercial outcomes

If ΔKL measures something commercially real, it should predict outcomes that cost firms money. It does, and the pattern splits cleanly by the level of observation. At the *mark* level, the trademark lifecycle selects against forward-leaning vocabulary: those applications register slightly less often, and the marks that do register fail the first maintenance gate more often. At the *firm* level, the same signal points the other way: firms whose filings lean forward are more likely to eventually go public, earn higher gross margins, and earn excess stock returns. The two layers together read as a risk-and-return profile. Forward-leaning vocabulary marks a commercial bet; the bets die more often as individual marks, and the firms that place them do better.

3.1 Mark-level selection: registration and the first maintenance gate

Outcome definitions. Outcomes derive from the USPTO status-code dictionary. A filing reached registration if its registration date is populated. A registered mark faces the Section 8 use-declaration gate in years five to six; *Cancelled–Section 8* (status 710) marks failure, and *Section 8 accepted* (701/702, partial variants 703–705) or subsequent renewal (800) mark survival. Because older registrations face repeated gates, the gate analysis uses the 2016–2018 registration cohort, for which exactly one Section 8 gate has elapsed by the 2025 data snapshot.

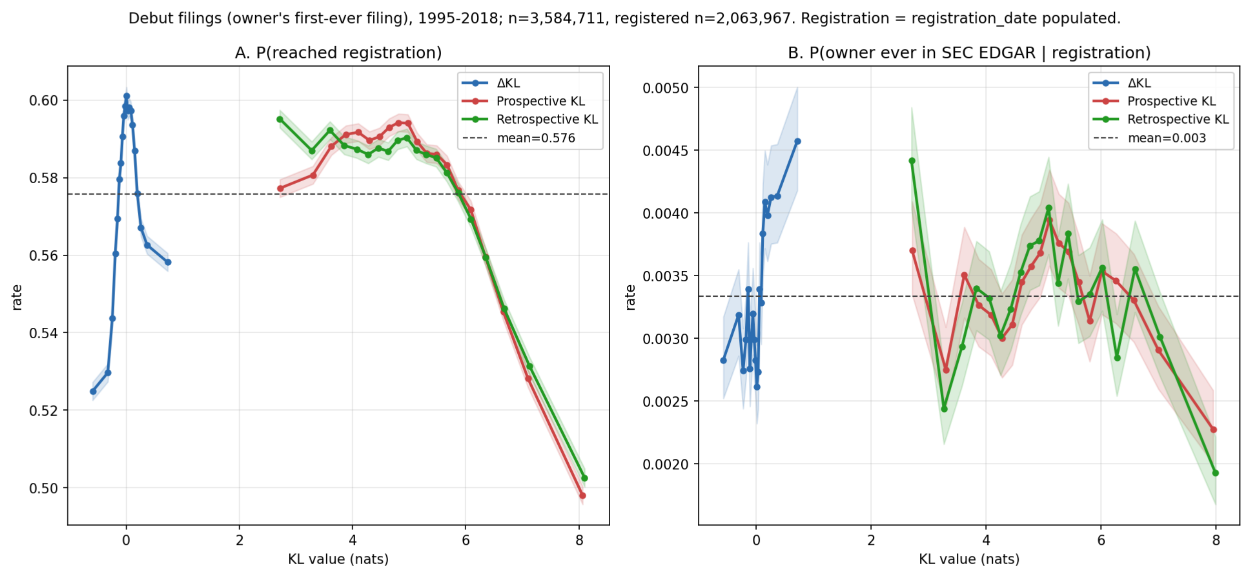

Figure 3: Outcomes for debut filings (owner's first-ever filing) across all 45 NICE classes, filing years 1995–2018, in quantile bins with 95% intervals. $n = 3{,}584{,}711$; registered subset $n = 2{,}063{,}967$. *A:* P(reached registration). *B:* P(owner ever in SEC EDGAR | registration).

Registration (Figure 3A; 3.58 million debut filings, 57.6% base rate) is an inverse-U in ΔKL: the flux-neutral middle registers best (59.8% at the third quintile) and both tails register less (54.0% at the bottom quintile, 56.6% at the top). In the KL level the top quintile is penalized on both axes (53.3% for the most past-unusual fifth, 53.5% for the most future-unusual), so the most distinctive applications clear examination least often. The inverse-U is the rule, not a pooled artifact: estimated class by class on all filings 1995–2018, the middle-versus-tails depth is positive in 37 of 44 classes with sufficient volume and statistically distinguishable from zero in 33 (the per-class figure and table are in the repository).

The registration inverse-U is also why analyses that skip the registration conditioning see an inverse-U in downstream outcomes. Any unconditional success measure, mark survival included, is the product of the registration rate and the conditional rate, and the registration factor stamps its middle-favoring shape onto the composite: the flux-neutral middle registers best, so it appears to "survive" best unconditionally even where the conditional gate pattern says otherwise. Appendix A reproduces the outcome analysis without the registration conditioning to make the composite shape visible.

The first Section 8 gate is where forward-leaning marks pay. Figure 4 shows the pooled result on 765,154 gated registrations: survival falls across ΔKL quintiles from 49.2% to 43.3%, a −5.9pp lift (SE 0.2pp), while it *rises* in the raw KL level (+6.5pp prospective, +7.6pp retrospective). Distinctive text survives the use requirement better; forward-leaning text survives it worse. A natural mechanism runs through the use declaration itself: Section 8 requires continued use in commerce, and vocabulary that bets on where the category is going belongs disproportionately to products that had not yet materialized at filing and sometimes never do.

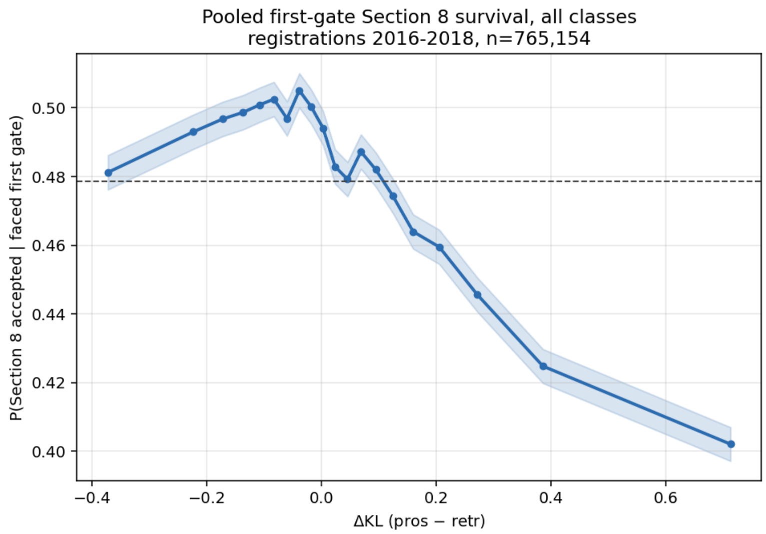

Figure 4: First Section 8 gate survival by ΔKL, pooled across classes. Registrations 2016–2018; survival = status 701/702/703/704/705/800, failure = status 710.

The gate result is broad but not universal. Per class (Figure 5), the ΔKL top-versus-bottom quintile lift is negative in 31 of 41 classes with sufficient gated volume, with the largest negative lifts in physical-goods classes (medical apparatus −21.7pp, tobacco −24.1pp, lighting −19.9pp, machinery −18.4pp) and small positive lifts concentrated in services (legal services +7.4pp, telecommunications +6.3pp). One reading is that a forward-leaning physical-goods filing requires a product to exist by the gate year while a forward-leaning services filing can be serviced with capacity on hand; testing the gradient against class-level production lead times is a follow-up computation.

Summarizing the cross-industry picture: the registration inverse-U is near-universal (37 of 44 classes), while the gate penalty is the majority pattern with a coherent minority. The gate exceptions are not noise scattered across the forest; they cluster in service classes where the cost of making a forward-leaning filing true by the gate year is lowest. One small class, lace and embroidery (026), shows a genuine mirror image, with tails registering better and forward-leaning marks passing the gate more often; every other class fits the headline pattern on at least one of the two steps.

3.2 Firm-level outcomes: listing, margins, returns

Among registered debut filings, the probability that the owner ever appears in the SEC EDGAR listed universe (Figure 3B) rises in ΔKL from 0.29% at the bottom quintile to 0.42% at the top, roughly a 40% relative lift on a 0.33% base. The KL levels carry no such signal (flat to slightly negative across quintiles); as with returns below, the information is in the signed resonance, not in novelty per se. The debut restriction matters: at the whole-firm level, selection into public markets is dominated by filing volume (a tenfold increase in filings raises the odds of listing roughly fivefold), and in the all-filings pool the relationship inverts, the line-extension confound in raw form. The construct's listing signal is concentrated at the moment of entry, before portfolio-scale effects swamp it.

Margins and returns on the SEC panel

On a panel of 5,319 firm-years (1,597 SEC-matched trademark owners, 2009–2024), trailing 3-year mean ΔKL predicts gross margin at +3.2pp per σ ($t = 5.3$), controlling log revenue and R&D intensity. The decomposition is instructive: prospective surprise enters at +14.6pp and retrospective surprise at −19.1pp (both $|t| > 3.5$), so the margin premium attaches to vocabulary that is unusual against the past *and* close to the future, exactly the resonance configuration. On first-time filers alone, where reverse causality from established success is weakest, the effect remains at +1.7pp per σ ($n = 6{,}525$ firms).

Excess stock returns over the S&P 500 show the same signature. Firm-year mean ΔKL predicts +1.5pp per σ at the 1-year horizon ($t = 3.0$), +3.5pp at 2 years, +5.0pp at 3 years, +4.1pp at 4 years, and decays to insignificance at 5 years. Prospective surprise alone predicts approximately zero at every horizon. Mere novelty at filing time carries no return information; novelty the future moves toward does. A larger headline figure from a debut-only specification (+28pp at 4 years) does not survive a sample-size diagnostic and is not claimed; the firm-year profile is the defensible one.

4 Distinctness from patents

If ΔKL and patent counts were noisy measures of the same underlying construct, they should correlate within firms over time. They do not.

The within-firm analysis uses a published panel of 174,569 firm-years of trademark owners name-matched to PatentsView patent assignees (PatentsView, 2024); the fixed-effects estimation uses the 94,106 firm-years across 14,470 firms with within-firm variation. The panel and matching code are in the public repository. The matching normalizes both sides (strip corporate-form suffixes and punctuation, lowercase) and joins on exact normalized-name equality. One provenance note: the published panel was constructed under an earlier exponential-decay kernel variant of the construct; its firm-year mean ΔKL correlates at $r = 0.87$ with the current flat-window scoring on 68,013 matched owner-years, so the results below are expected to be kernel-stable, and re-estimation on the current scoring is owed with the next panel release. I fit a two-way (firm + year) fixed-effects regression of standardized firm-year mean ΔKL on log(1 + patents) with firm-clustered standard errors.

The coefficient is negative and precisely estimated, strengthening under filing-count control. The strongest negative coefficient sits on patents$_{t-1}$: in years after a firm patents heavily, its trademark vocabulary regresses toward class-typical. A reading consistent with the data is temporal substitution between patent prosecution and market-facing brand work; either way, the two measures are not interchangeable.

The contrast with expert judgment completes the discriminant picture. Across the 7,431-firm public panel, firms appearing on the BCG Most Innovative lists (2021–23) sit +0.28σ above the panel mean in ΔKL ($n = 31$ matched, $p = 0.012$), firms on the MIT Technology Review 50 Smartest Companies list (2015) sit +0.50σ above ($n = 17$, $p = 0.0006$), and the pooled union sits +0.34σ above ($n = 43$, $p = 0.0003$). Patent counts are uncorrelated with ΔKL on the same panel (Pearson

Table 1: Within-firm ΔKL versus patenting. Coefficients in standard deviations of ΔKL per unit $\log(1 + \text{patents})$.

Specification	coef (σ)	SE	t
Between-firm correlation (firm means)	−0.020	—	—
Within-firm correlation (firm+year demeaned)	−0.015	—	—
Two-way FE, contemporaneous	−0.048	0.006	−7.9
robustness: patents = max over matched assignees	−0.048	0.006	−8.0
controlling log(n_filings)	−0.051	0.006	−8.4
Lead/lag: patents_{t-1}	−0.054	0.009	−5.8
patents_{t}	−0.019	0.010	−1.9
patents_{t+1}	+0.010	0.010	+1.0

+0.010). The expert lists reward something patents do not capture, and ΔKL tracks it. The matched samples are small and the result is a discriminant check, not a headline.

5 Application: cross-industry diffusion, measured directly

Because ΔKL is computed from open text, the same corpus supports measuring where business vocabulary is born and how it travels. This section demonstrates the construct's reach beyond firm-level validation.

5.1 Phrase-level transit

For 14 curated business-model phrases, I record the first year each of the four classes crosses a 1% prevalence floor among granted filings.

Table 2: First year a theme reaches 1% prevalence among granted filings, by class. "–" = never clears the floor.

Theme	software	tech-svcs	transport	advert/retail
as-a-service	2011	**2009**	2013	2012
mobile-app	2010	**2009**	2012	2012
cloud	2012	**2010**	2011	2014
streaming	**2008**	2008	2013	2014
artificial-intelligence	2017	**2016**	2018	2018
augmented/virtual-reality	**2015**	2016	2022	2022
blockchain / crypto	2021	**2018**	–	2022
real-time	**2001**	2003	2009	2014
on-demand	2017	2016	**2014**	2017

Software- and tech-services-origin phrases arrive in the physical-economy classes with lags of one to thirteen years; the only counterexample in the set is on-demand, which originates in transport. The 1% floor is sensitive to class size (transport has roughly one-eighth the volume of software), and the asymmetry is a four-class result that a 45-class build would test properly; the current evidence is consistent with the directional reading but cannot exclude a corpus-construction alternative.

5.2 Theme-level structure

A theme-level analysis with unsupervised topic extraction corroborates the phrase-level pattern without depending on which phrases were picked. I fit LDA with $T = 50$ on a stratified 200,000-filing sample (vocabulary $V = 69{,}361$) and transform across the full 2.32M-filing pool. The resulting topics are recognizable business-model components (health/wellness; autonomous loading and material handling; retail store services; electronic transmission and computer networks; vehicles and traffic).

For each (theme, class) cell that reaches the 1% floor and is tracked at least six years, I fit the Bass diffusion model (Bass, 1969). Of 118 retained fits ($R^2 \geq 0.7$), the median innovation coefficient p is 0.018, the median imitation coefficient q is 0.110, and median $q/p \approx 6.5$, in the range typically reported for consumer-product adoption. Most fits are platform-like; a small tail is fad-shaped (renewable-energy and virtual-reality themes show $q/p > 40$, with steep adoption and visible recession).

Tagging each multi-class theme's origin class (earliest to cross the floor) and counting one directed edge from the origin to each later arrival class gives 72 origin-to-arrival edges over the 33 multi-class themes. Software (009) is the dominant origin, contributing 33 of the 72 edges against the 18 expected under symmetry, and receiving only 10; the software-plus-tech-services pair sends 26 edges to transport and advertising/retail and receives 18 back. The same asymmetry reproduces under independently-fitted NMF on the same sample and vocabulary, with seven of ten curated seed phrases recovering a clear best topic. Both methods are bag-of-words factorizations, so this robustness is within-paradigm; a sentence-embedding or graph-community extraction is the natural cross-paradigm test and is not run here.

6 Do new themes come from incumbents or entrants?

The corpus invites population-share questions, and it punishes them when the base rate is ignored. As a worked example with substantive interest of its own: when a new theme appears, is it carried by new entrants or by incumbents diversifying in? Schumpeter (1942) introduced the distinction (Mark I entrant-led, Mark II incumbent-led), and the standard empirical form reports the entrant share among early adopters.

Applying the standard form to the 50 LDA themes, the 50 earliest carriers per theme are 73.7% entrants on average, and 48 of 50 themes are majority-entrant. The raw numbers look like strong entrant-led adoption. But the corpus's debut rate among all granted filings in this period is 76.4% under year-matched weighting; most filings are by owners debuting that year. Against that baseline the theme-carrier excess is -2.7pp, and 28 of 50 themes sit below baseline. Themes inherit the corpus's high debut rate without adding to it.

The residual pattern is the interesting part. The themes carried disproportionately by incumbents relative to baseline include environmental/sustainability, natural gas and petroleum, artificial intelligence, and cloud computing. The canonically new technology themes of the 2010s and 2020s lean incumbent: established firms diversifying into AI and cloud pull the vocabulary signal forward at least as much as de-novo entrants do. This is consistent with the within-firm patent contrast of Section 4; the firms moving the vocabulary are not necessarily the firms moving the patents.

7 A research agenda the construct opens

Each of the following is computable from the public corpus and panels with the apparatus of this paper. None is run to completion here.

Are transferred ideas as valuable away from home? The flow graph identifies, for every theme, its origin class and arrival classes. Linking carrier filings to the SEC panel tests whether a theme earns the same margin and survival premia in destination industries as in its origin. A first-pass within-firm estimate finds a contemporaneous gross-margin lift of +1.4pp per σ of borrowed novelty ($t = 2.1$) that fades within a year; whether portability is the rule or the exception is open.

Suppliers versus deployers. When a phrase crosses industries, the carrier filings split into firms whose offer *is* the new thing and firms whose offer *supplies* the firms doing the new thing, the sellers of pick-axes rather than the panners for gold. The goods/services text is rich enough to separate the two roles, and the gold-rush intuition that suppliers outperform deployers is testable against renewal, margins, and listing.

Demographic concentration and where themes are born. Surname-imputed demographic composition of filing owners, applied to US-domiciled individual filers, identifies industry enclaves in which a demographic group is heavily over-represented. Preliminary work finds enclave filings sit on the innovation tail of ΔKL rather than the harvest tail, and that owners who diversify across classes list publicly at many times the single-class rate. Whether enclave industries disproportionately originate themes, and whether enclave-rooted firms carry them outward, is a direct join of the demographic panel to the flow graph.

External codification. The USPTO ID Manual periodically standardizes acceptable goods/services wording. If high-ΔKL filings use terms the Manual later codifies, while those terms are still rare in class, the construct anticipates institutional recognition, the strongest available external validation and one that requires only the Manual's version history.

Cross-paradigm theme extraction. The diffusion structure has been shown method-stable within bag-of-words factorizations. A sentence-embedding clustering or graph-community extraction on the same corpus would establish (or refute) paradigm independence.

8 Scope and limits

The paper makes four substantive claims. ΔKL, computed as in Barron et al. (2018); Murdock et al. (2017) on granted USPTO trademark filings, produces interpretable structure on commercially-incentivised legal text. It marks commercial risk-taking, with opposite signs at the mark and firm levels: forward-leaning marks register slightly less often and fail the first Section 8 maintenance gate more often (−5.9pp across ΔKL quintiles pooled; negative in 31 of 41 industries), while the firms filing them show higher gross margins, 1–4 year excess returns, and a higher eventual-listing rate among first-time filers, with the signed resonance carrying the firm-level signal and prospective surprise alone carrying none. It is empirically distinct from patent-track invention within firm while aligning with expert innovation judgment. And it supports direct measurement of cross-industry vocabulary diffusion, with multi-year origin-to-arrival lags, Bass-fittable adoption, and strongly asymmetric flow out of software.

The paper rejects two readings the construct does not support. ΔKL does not measure innovation in any strong sense; the defended primitive is lexical resonance on commercial text, and the validation evidence bounds the interpretation. ΔKL does not identify an exploitable extreme-tail return premium; the large debut-specification figure fails a sample-size diagnostic.

The Section 8 gate result is identified on a single registration cohort (2016–2018, the cohort for which exactly one gate has elapsed), so cohort-specific shocks, including the pandemic years falling inside the gate window, are not separable from the ΔKL gradient; replication on the 2019–2021 cohort as its gate completes is the direct check. The margin and returns results are associational, identified within an SEC-matched panel that itself selects on scale. The theme-level diffusion analysis is a four-class build; per-class ΔKL scores now exist for all 45 classes, but the theme panel and flow graph have not been re-estimated on the full set. The expert-list discriminant rests on 43 matched firms.

Questions owed to follow-up work: the four agenda items of Section 7; whether the mark-level gate penalty reflects vocabulary position itself or selection on owner covariates; whether the cross-industry gradient in the gate penalty tracks production lead times or vocabulary churn; and re-estimation of the outcome analyses under topic-distribution scoring.

9 Reproducibility

Code, firm-year panels, and analysis outputs are available at `https://github.com/ericsilver/novelty`. The pipeline rebuilds from a USPTO Open Data Portal API key with a single `make tm` command (the underlying TRTYRAP backfile is approximately 12 GB streamed once). The published panels include the SEC-matched firm-year ΔKL panel ($n = 59{,}033$) and the PatentsView-joined panel ($n = 174{,}569$). Analysis scripts and their outputs are indexed in the repository `README.md`.

A The outcome analysis without registration conditioning

Figure 6 repeats the lifecycle outcome analysis without the registration conditioning, on filings 2013–2015 pooled across all classes (a window whose registrations land by ∼2017 and face at least one Section 8 gate by the 2025 snapshot; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two gates). Panel A is the unconditional composite, the probability that a filing both registers and passes the gate, measured over all filings including the never-registered. It is an inverse-U in ΔKL (quintiles 22.0%, 22.9%, 23.0%, 22.5%, 21.4%; middle-versus-tails depth +1.4pp, SE 0.1pp), because the composite inherits the registration step's middle-favoring shape from Figure 3A. Unconditional survival analyses on this corpus therefore recover an inverse-U whether or not the maintenance gate itself favors the middle, which is why the body of the paper conditions on registration throughout. Panel B shows the conditional gate outcome on the same filings for contrast; on this mixed one-to-two-gate cohort it is mildly inverse-U with a clearly penalized forward tail, consistent in the right tail with the cleaner single-gate cohort of Figure 4 and a reminder that the left tail of the gate result is cohort-sensitive.

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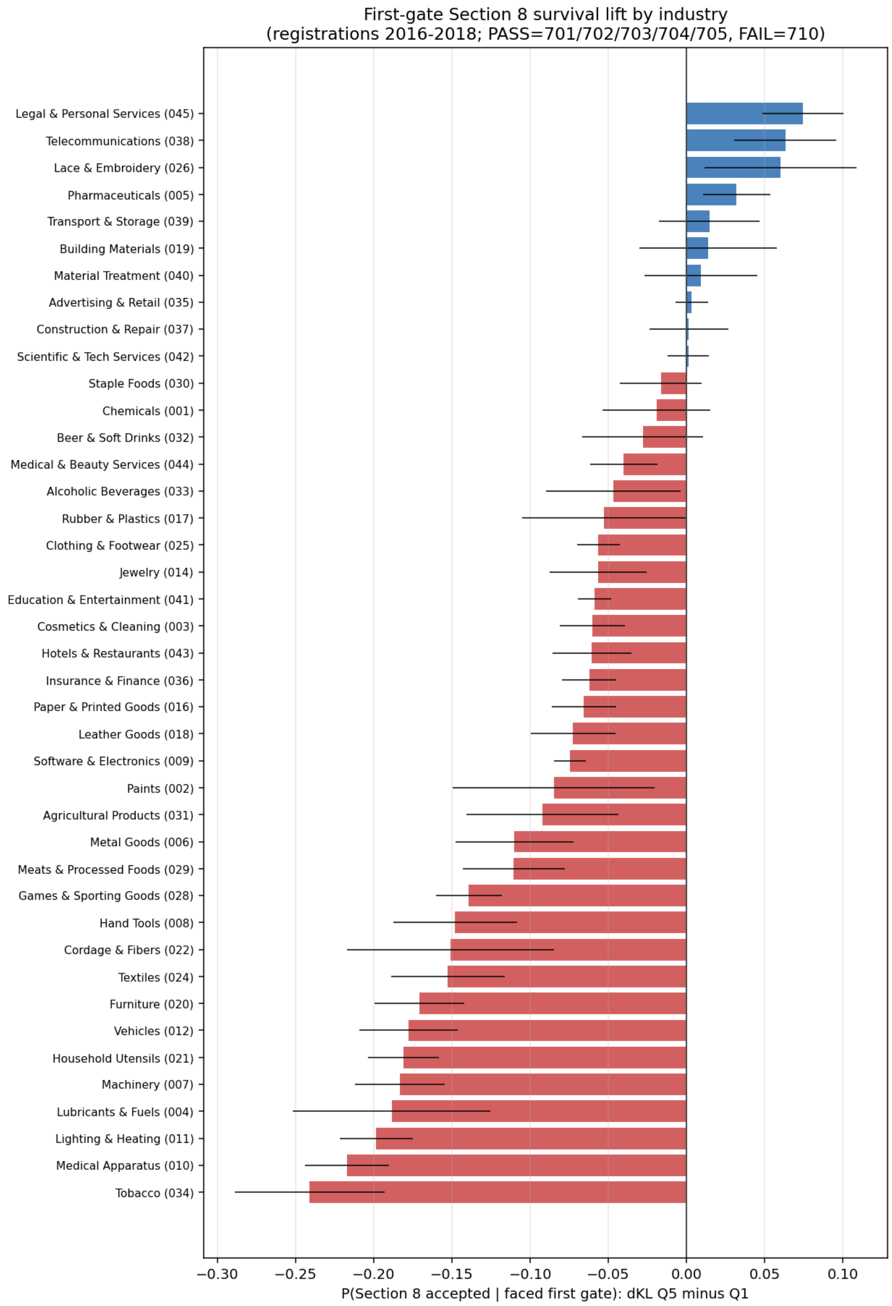

Figure 5: First-gate Section 8 survival lift (ΔKL top minus bottom quintile) by industry, registrations 2016–2018, with 95% intervals.

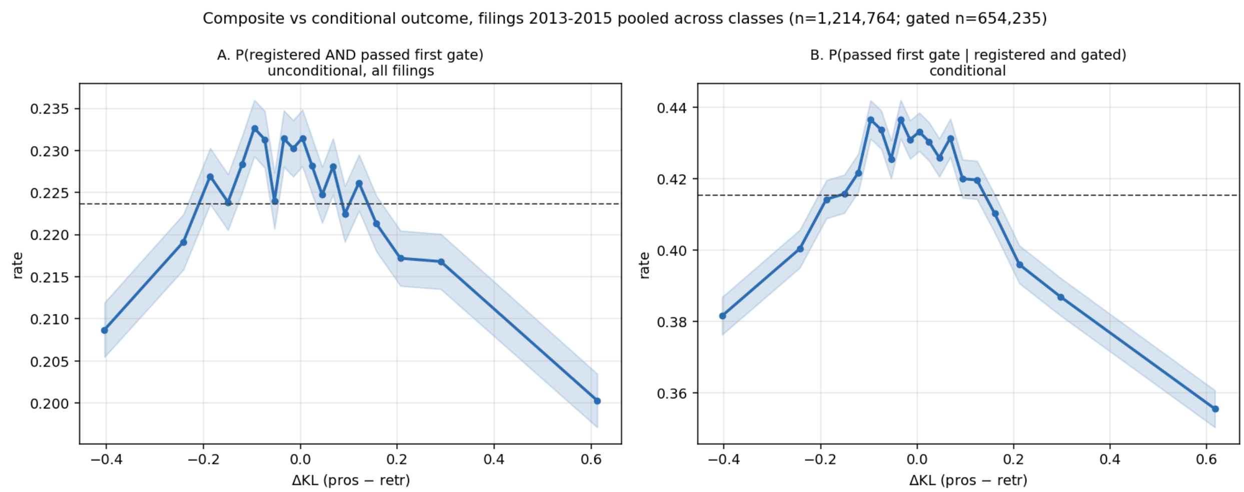

Figure 6: Filings 2013–2015 pooled across classes (n = 1,214,764; gated subset n = 654,235). *A:* the unconditional composite P(registered and passed first gate) over all filings. *B:* P(passed | registered and gated) on the same filings.